

ELECTRONICS



THE ELECTRONICS REVOLUTION is rapidly changing our world: whether we are at home, at work, or out shopping, we are surrounded by electronic machines and equipment. Electronics involves using devices called components to control electric currents, which are flows of tiny, electrically charged particles of matter called electrons. An electronic circuit is an arrangement of linked components – such as transistors and diodes – that manipulates current in order to carry out a specific task, such as adding numbers in a calculator.

Light Emitting Diodes (LEDs) glow when current passes through them, and are used to indicate that a device's power supply is on.

Variable resistors allow the level of current flowing through a circuit to be adjusted.

Capacitors are components that store electric charge; electrolytic capacitors can store more charge than ceramic ones.

Radio circuit board and components

Electrolytic capacitors

Ceramic capacitors

Integrated circuits consist of a plastic case containing a complete circuit etched on to a tiny silicon chip.

Power cables

Resistors allow only a fixed amount of electric current to flow through a circuit.

Circuit board

The components for an electronic device, such as a radio, are attached to a circuit board, which is a flat base with metal tracks running along its underside. The components are secured to the tracks using an alloy called solder. The tracks link the components to form a circuit.

Inductors are wire coils that produce magnetic fields when current passes through them, creating a resistance that restricts the flow of current.

Variable capacitors can be adjusted to store varying levels of charge; in radios, they are used to select radio stations.

Diodes allow electric current to pass through them in one direction only.

Transistors can be used to amplify electrical signals (make them stronger) or switch circuits rapidly on and off.

Semiconductors

The element silicon is a type of material called a semiconductor, because it conducts electricity only under certain conditions. The properties of a semiconductor can be altered by adding chemical impurities to it in a process called doping. Doped semiconductors are used to make diodes, transistors, and many other electronic components.

Silicon crystal

Pins connect to circuit board.

Microprocessors
Many electronic devices – including computers – are controlled by circuits called microprocessors, or "silicon chips". A microprocessor is made from a single slice of doped semiconductor. The circuit, which may contain thousands of components, can carry out many complex tasks.

Silicon chip sealed under metal cover

Ceramic casing

Uses of electronics

Electronic circuits are either analogue or digital. Analogue circuits deal with continuously varying electric currents, such as television and radio signals. Digital circuits process information in the form of thousands of on-off pulses of electric current every second.

Pocket calculator

Video-game console

CD-ROM drive

Liquid crystal display

Handset with control keys

Calculator

A calculator's digital circuits split up a calculation into a series of simple steps, each of which is performed at high speed.

Video game console

Digital circuits inside the console control the play. The console sends an analogue signal to a TV screen, which displays a picture of the game.

William Shockley

US physicist William Shockley (1910–89) was part of a three-man team that invented the transistor in 1947. The transistor made it possible to build tiny electronic circuits and so develop more compact electronic devices.



Remote control

Pressing a button on the remote-control of a TV – for example, to change channels – makes an LED flash pulses of infrared light to the TV set. The TV set decodes the pulses and obeys the instruction.

Television controls

Liquid crystal display

Timer and clock controls

Number keypad

Video recorder controls

TV/video remote control

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